

11 Poisson Distributions

- Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of “relatively rare” events that occur over a certain interval of time/in a certain location.

Example 11.1. Recall the matching problem from Example 6.3, Exercise 7.3, Exercise 8.2, and Exercise 9.1. There are n distinct objects that are shuffled and placed uniformly at random in n distinct spots with one object per spot, and X is the number of correct matches of object with spot. Note: n is a fixed and known number, but we are considering different values of it.

We have seen that $E(X) = 1$ for any n and that unless n is small (e.g., $n \leq 6$), $\text{Var}(X) = 1$ and the approximate distribution of X is given by Table 11.1.

Starting with $x = 0$, describe the distribution of X in terms of relative likelihoods:

1. 1 is [blank] times as likely as 0
2. 2 is [blank] times as likely as 1
3. 3 is [blank] times as likely as 2
4. 4 is [blank] times as likely as 3

Table 11.1: Poisson(1) distribution, the approximate distribution of X , the number of matches in the matching problem for general n and uniformly random placement of objects in spots.

x	P(X=x)
0	0.36788
1	0.36788
2	0.18394
3	0.06131
4	0.01533
5	0.00307
6	0.00051
7	0.00007

5. 5 is [blank] times as likely as 4

6. 6 is [blank] times as likely as 5

7. In general, x is [blank] times as likely as $x - 1$, for $x = 1, 2, 3, \dots$

- A random variable X has a **Poisson distribution** with parameter $\mu > 0$ if the possible values are $0, 1, 2, \dots$ and the pmf follows the

$$\text{Poisson}(\mu) \text{ pattern: } x \text{ is } \frac{\mu}{x} \text{ as likely as } x - 1 \quad (\text{for } x = 1, 2, 3, \dots)$$

- As a function of $x = 0, 1, 2, \dots$, a $\text{Poisson}(\mu)$ pmf increases for $x < \mu$ and then decreases for $x > \mu$
 - If μ is a whole number, then the Poisson pmf assigns the same probability to the values μ and $\mu - 1$.
 - If $\mu < 1$ then $x = 0$ has the highest probability and the pmf decreases from there.

- If X has a $\text{Poisson}(\mu)$ distribution then

$$E(X) = \mu$$

$$\text{Var}(X) = \mu$$

Example 11.2. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Technically, there is no fixed upper bound on what X can be, so mathematically it is convenient to consider $0, 1, 2, \dots$ as the possible values of X . Assume that X has a Poisson distribution with parameter 2.4.

1. Interpret 2.4 in context.
2. Identify what the $\text{Poisson}(2.4)$ pattern implies in context.
3. Sketch a plot of the distribution of X .
4. Assuming that the probability that X is greater than 9 is negligible, make a table representing the distribution of X .
5. The pmf of X is

$$P(X = x) = e^{-2.4} \frac{2.4^x}{x!}, \quad x = 0, 1, 2, \dots$$

Use the pmf to compute $P(X = 0)$.

6. Use the pmf to compute $P(X = 1)$.
 7. Use the pmf to compute $P(X = 2)$.
 8. Use the pmf to construct a table and plot representing the distribution of X . Does it match your previous work?
 9. Use the pmf to compute $E(X)$ and verify that it is 2.4.
 10. Compute and interpret $P(X \leq 2)$.
-
- A discrete random variable X has a Poisson distribution with parameter $\mu > 0$ if and

only if its probability mass function p_X satisfies

$$p_X(x) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

- The Poisson pattern implies that the shape of a Poisson pmf as a function of x is given by $\mu^x/x!$.

$$p(x) = p(0) \frac{\mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

- The constant $p(0) = e^{-\mu}$ simply renormalizes the pmf so that the probabilities sum¹ to 1.

Example 11.3. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Assume that X has a Poisson(2.4) distribution.

1. In what ways is this like the Binomial situation?

2. In what ways is this NOT like the Binomial situation?

3. In what sense are home runs “relatively rare”?

- Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of “relatively rare” events that occur over a certain interval of time/in a certain location
- **Poisson paradigm.** Let $A_1, A_2, \dots, A_n$ be a collection of n events. Suppose event i occurs with marginal probability $p_i = P(A_i)$. Let $N = I_{A_1} + I_{A_2} + \dots + I_{A_n}$ be the random variable which counts the number of the events in the collection which occur. Suppose

– n is “large” (but not necessarily fixed),

¹Recall the series expansion $e^\mu = \sum_{x=0}^{\infty} \frac{\mu^x}{x!}$

- $p_1, \dots, p_n$ are “comparably small” (but not necessarily the same), and
- the events $A_1, \dots, A_n$ are “not too dependent” (but not necessarily independent).
- Then N has an approximate Poisson distribution with parameter $E(N) = \sum_{i=1}^n p_i$.

11.1 Exercises

Exercise 11.1. Continuing Example 11.1. Explain why X in the matching problem has a Poisson distribution when n is large.

Exercise 11.2. Suppose that the number of fatal crashes in SLO County in a week follows, approximately, a $\text{Poisson}(0.53)$ distribution, independently from week to week. Let X be the number of fatal crashes in a period of 4 consecutive weeks (close enough to a “month”).

1. Make an educated guess for $E(X)$.
2. Compute $P(X = 0)$.
3. Compute $P(X = 1)$.
4. How could you use simulation to approximate the distribution of X ?

5. Use simulation to approximate the distribution of X . Does the approximate distribution follow (approximately) the Poisson pattern?

 6. Use a Poisson pmf to compute $P(X = 5)$.
-
- **Poisson aggregation.** If X and Y are independent, X has a $\text{Poisson}(\mu_X)$ distribution, and Y has a $\text{Poisson}(\mu_Y)$ distribution, then $X + Y$ has a $\text{Poisson}(\mu_X + \mu_Y)$ distribution.
 - If component counts are independent and each has a Poisson distribution, then the total count also has a Poisson distribution.